Ejercicio 4.3.7

Tuesday, September 30, 2025 12:04 AM

2.

a) $If\rangle_t \leftrightarrow f(t) = 5t+3t^2+0t^3$.

$\mathcal{B}_m = \{1, t, t^2, t^3, t^4\}$ $\quad B_2 = \{P_0, P_1, P_2, P_3, P_4\}$

$B_2 = \{1,\ t,\ \frac{1}{2}(3t^2-1),\ \frac{1}{2}(5t^3-3t),\ \frac{1}{8}(35t^4-30t^2+3)\}$

$[f]_{\mathcal{B}_m} = \begin{pmatrix} 0 \\ 5 \\ 3 \\ 0 \\ 0 \end{pmatrix}$ $\quad f(t) = \sum_{k=0}^{4} c_k P_k(t)$

$\{P_k\}$ es ortogonal, $c_k = \frac{\langle f, P_k\rangle}{\langle P_k, P_k\rangle}$; $c_0 = \frac{\langle f, P_0\rangle}{\langle P_0, P_0\rangle} = 1$

$[C]_{B_2} = \begin{pmatrix} 1 \\ \frac{37}{5} \\ 2 \\ \frac{8}{5} \\ 0 \end{pmatrix}$ $\quad f(t) = 1\cdot P_0(t) + \frac{37}{5} P_1(t) + 2P_2(t) + \frac{8}{5} P_3(t)$

$$L = \begin{pmatrix} 1 & 0 & -\frac{1}{2} & 0 & 3/8 \\ 0 & 1 & 0 & -3/2 & 0 \\ 0 & 0 & 3/2 & 0 & -15/4 \\ 0 & 0 & 0 & 5/2 & 0 \\ 0 & 0 & 0 & 0 & 35/8 \end{pmatrix}$$

$\mathcal{B}_m \to \mathcal{B}_2$ $\quad S = L^{-1}$

$$S = L^{-1} = \begin{pmatrix} 1 & 0 & 1/3 & 0 & 1/5 \\ 0 & 1 & 0 & 3/5 & 0 \\ 0 & 0 & 2/3 & 0 & 4/7 \\ 0 & 0 & 0 & 2/5 & 0 \\ 0 & 0 & 0 & 0 & 8/35 \end{pmatrix}$$

$[C]_{B_2} = S[f]_{\mathcal{B}_m}$ da el vector $(1, \frac{37}{5}, 2, \frac{8}{5}, 0)^T$

(b) $P_{[-1,1]2}$

$P_V(f) = \sum_{k=0}^{2} \frac{\langle f, P_k\rangle}{\langle P_k, P_k\rangle} P_k$ $\quad c_0 = 1,\ c_1 = \frac{37}{5}$ y $c_2 = 2$

$P_V(f) = 1\cdot P_0 + \frac{37}{5} P_1 + 2P_2$

$P_V(f)(t) = \frac{37}{5}t + 3t^2$

$P_{mat} = \mathcal{B}(\mathcal{B}^T G \mathcal{B})^{-1}\mathcal{B}^T G$.

c) Operador inverso T^{-1}. el adjunto del operador $T^\dagger$ y precisa si T. es hermitiano o unitario

$D = \frac{d}{dt}$ $\quad T = e^D$ $\Rightarrow$ $(Tf)(t) = e^D f(t) = f(t+1)$

$T^{-1} = e^{-D}$ $\quad (T^{-1}f)(t) = f(t-1)$

$T^\dagger \langle \cdot, \cdot\rangle = \int_{-1}^{1} \cdots$ satisface

$\langle Tf, g\rangle = \langle f, T^\dagger g\rangle \ \forall\ f, g$.

$$= \int_{-1}^{1} f(t+1) g(t)\, dt = \int_0^2 f(u) g(u-1)\, du$$

$\int_{-1}^{1} f(u)(T^\dagger g)(u)\, du$ $\quad (T^\dagger g)(u) = \begin{cases} g(u-1) & u \in [0,1] \\ 0, & u \in [-1,0) \end{cases}$

no existe una formula simple de la forma $T^\dagger f = f(\cdot + a)$ con a constante que pertenezca en $P_{[-1,1]}$.

$T \neq T^\dagger$

$T^\dagger \neq T^{-1}$

d) tenemos $[T]_{\mathcal{B}_m}$, ahora

$$[T]_{B_2} = L^{-1}[T]_{\mathcal{B}_m} L = \begin{pmatrix} 1 & 1 & 3/2 & 7/2 & 75/8 \\ 0 & 1 & 3 & 15/2 & 41/2 \\ 0 & 0 & 1 & 5 & 35/2 \\ 0 & 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$\mathrm{tr}(T) = 5$

$\det(T) = 1$

e) $[T^{-1}]_{\mathcal{B}_m}$ y tenemos $[T]_{B_2} = L^{-1}[T]_{\mathcal{B}_m} L$

$[T^\dagger] = G^{-1}[T]^T G$ $\quad b_{ij} = \langle b_i, b_j\rangle$

$G_{B_2} = \mathrm{diag}(\|P_0\|^2, \ldots, \|P_4\|^2) = \mathrm{diag}(2, 2/3, 2/5, 2/7, 2/9)$

se obtiene: $$[T^\dagger]_{B_2} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 & 0 \\ 15/2 & 5 & 1 & 0 & 0 \\ 49/2 & 35/2 & 7 & 1 & 0 \\ 63/2 & 123/2 & 63/2 & 9 & 1 \end{pmatrix}$$

En la base monomial, el gram $G_{\mathcal{B}_m} \int_{-1}^{1} t^{i+j}\, dt$

$$G_{\mathcal{B}_m} = \begin{pmatrix} 2 & 0 & 2/3 & 0 & 2/5 \\ 0 & 2/3 & 0 & 2/5 & 0 \\ 2/3 & 0 & 2/5 & 0 & 2/7 \\ 0 & 2/5 & 0 & 2/7 & 0 \\ 2/5 & 0 & 2/7 & 0 & 2/9 \end{pmatrix}$$

T^{-1} y $T^\dagger$

$[T^{-1}]_{\mathcal{B}_2} = L^{-1}[T^{-1}]_{\mathcal{B}_m} L$ $\quad [T^\dagger]_{\mathcal{B}_2} = L^{-1}[T^\dagger]_{\mathcal{B}_m} L$.